

SOFTWARE REQUIREMENTS SPECIFICATION

AI – Powered Resume Screening And Personalized Interview Preparation Platform

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1.0.Introduction

The **AI-Powered Resume Screening and Personalized Preparation Platform** is a web-based application that streamlines the campus recruitment process for students, recruiters, and placement administrators. The system uses Artificial Intelligence to automate resume screening, generate ATS scores, match resumes with job descriptions, and assist students in interview preparation. It provides a centralized platform that improves recruitment efficiency, reduces manual effort, and enhances the overall placement experience.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the **AI-Powered Resume Screening and Personalized Preparation Platform**. This document serves as a guideline for the design, development, testing and deployment of the system. It also ensures a common understanding of the project among developers and faculty members.

1.2Scope of the Project

The proposed system provides a centralized platform for campus placements with separate modules for students, recruiters, and placement administrators.

The system includes:

- Student registration and profile management
- Resume upload and AI-based resume analysis
- ATS score generation and resume suggestions
- Resume and Job Description matching

- AI-generated interview questions and mock interview feedback
- Recruiter dashboard for candidate screening and ranking

The project will be developed as a responsive web application and deployed on the cloud.

1.3 Glossary

Term	Definition
AI	Technology that enables computers to perform tasks requiring human intelligence, such as resume analysis and interview question generation.
ATS Applicant Tracking System	A system used to evaluate and rank resumes based on predefined job requirements.
API Application Programming Interface	A web service architecture that enables communication between the frontend, backend, and external services.
JD	Job Description
Resume Parsing	Extracting information from resumes automatically
Match Score	AI-generated score indicating resume suitability
Sematic Matching	An AI technique that compares the meaning of a resume and job

	description rather than relying only on keyword matching.
Cloud Deployment	Hosting the application on cloud platforms such as Render or Vercel, making it accessible over the internet.
Admin	Placement Cell Administrator
JWT JSON Web Token	JSON Web Token is a secure authentication mechanism used for verifying and maintaining user sessions.
PostgreSQL	Relational Database Management System

1.4 References

The project is developed with reference to:

1. Smart India Hackathon (SIH) Problem Statement – Campus Placement and Internship Management System.
2. IEEE Software Requirements Specification (SRS) Guidelines.

2.0 Overall Description

The **AI-Powered Resume Screening and Personalized Preparation Platform** is a web based platform designed to simplify and automate the campus recruitment process using Artificial Intelligence. The system enables students to create

professional profiles, upload resumes, apply for jobs, and prepare for interviews, while recruiters can efficiently post job openings, screen candidates, and manage the recruitment process.

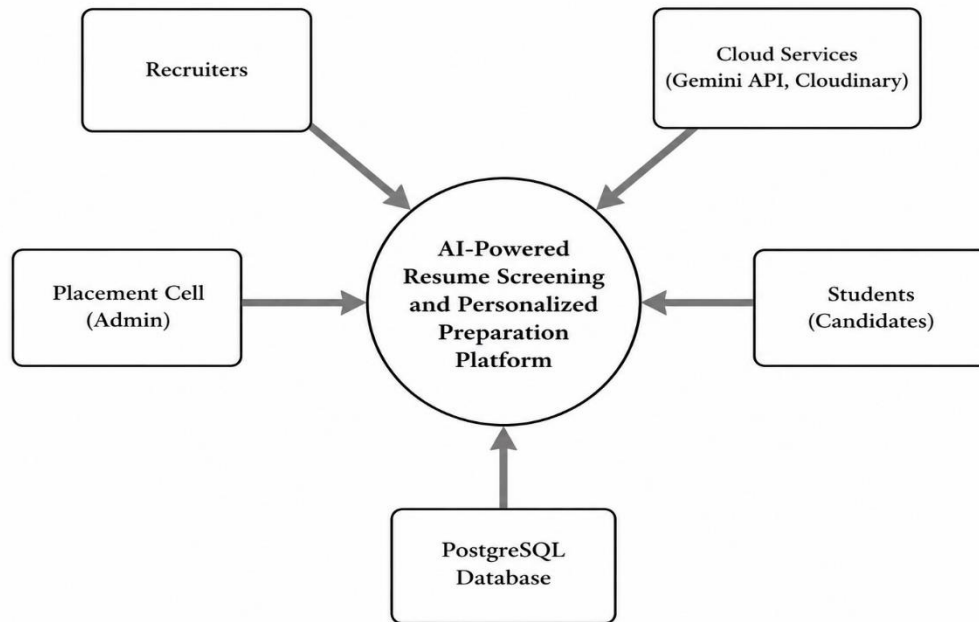
The system integrates AI technologies to analyze resumes, generate ATS scores, match resumes with job descriptions, identify skill gaps and provide interview preparation support. By centralizing recruitment activities into a single platform, the system reduces manual effort, improves candidate selection and enhances the overall campus placement experience.

2.1 Product Perspective

The AI-Powered Resume Screening and Personalized Preparation Platform is a standalone web application that serves as a centralized recruitment platform for educational institutions. It connects three primary stakeholders students, recruiters and placement administrators through a unified system.

The platform integrates cloud-based services for resume storage, relational databases for data management, and Artificial Intelligence for resume analysis, candidate-job matching and interview preparation. It is designed with a modular architecture, allowing future integration with external job portals, coding assessment platforms, and institutional management systems.

The platform will interface with:



2.2 Product Functions

The proposed system provides the following major functionalities:

- User Registration and Authentication
- Student Profile Management
- Resume Upload and Management
- AI Resume Parsing
- ATS Resume Score Generation
- Resume and Job Description Matching
- Job Posting and Management
- Job Search and Application
- Candidate Screening and Ranking

- AI Interview Question Generation
- Mock Interview and Feedback
- Dashboard and Placement Analytics
- User and Job Management by Administrator

2.3 User Classes and Characteristics

The system supports three categories of users.

Students (Candidates)

- ❑ Create and manage profiles
- ❑ Upload resumes
- ❑ Search and apply for jobs
- ❑ View ATS scores and resume suggestions
- ❑ Prepare for interviews using AI-generated questions

Recruiters

- ❑ Register company profiles
- ❑ Post and manage job openings
- ❑ Screen resumes
- ❑ Rank candidates
- ❑ Shortlist suitable applicants

Placement Administrator

- ❑ Manage students and recruiters
- ❑ Verify job postings

- ☐ Monitor placement activities
- ☐ Generate placement reports
- ☐ Maintain system records

2.3 Operating Environment

Software Environment

- ☐ React.js (Frontend)
- ☐ FastAPI (Backend)
- ☐ PostgreSQL (Database)
- ☐ Google Gemini API
- ☐ Cloudinary (Resume Storage)

Hardware Environment

- ☐ Desktop or Laptop
- ☐ Smartphone (Responsive Web Application)
- ☐ Internet Connection

Browser Support

- ☐ Google Chrome
- ☐ Mozilla Firefox
- ☐ Microsoft Edge

2.5 Design and Implementation Constraints

The development of the system is subject to the following constraints:

- The application will be developed as a web-based system.

- AI functionalities depend on the availability of the Google Gemini API.
- Stable internet connectivity is required for cloud deployment.
- The system will support only PDF and DOCX resume formats.
- The project will be completed within the Agile semester timeline.

2.6 User Documentation

The system will provide the following documentation:

- User Guide for Students
- Recruiter User Guide
- Administrator Guide
- Installation Guide
- API Documentation
- Frequently Asked Questions (FAQ)

2.7 Assumptions and Dependencies

Assumptions :-

- Users have basic knowledge of web applications.
- Recruiters provide accurate job descriptions.
- Students upload valid and updated resumes.
- Internet connectivity is available during system usage.

Dependencies :-

- ❑ Google Gemini API for AI functionalities.
- ❑ PostgreSQL database for data storage.
- ❑ Cloudinary for storing resume files.

🔗 Vercel and Render for deployment.

3.0 System Features and Functional Requirements

This section describes the functional requirements of the **AI-Powered Resume Screening and Personalized Preparation Platform**. The requirements are organized based on the major system modules. Each requirement is assigned a unique identifier (FR-XX) to ensure traceability throughout the software development lifecycle.

3.1 User Management and Authentication

3.1.1 Candidate Registration and Profile Management

ID	Requirement
FR-01	The system shall allow candidates to register using their email address and password.
FR-02	The system shall support secure user authentication through login and logout functionalities.
FR-03	The system shall allow candidates to reset their passwords using email verification.

FR-04	The system shall allow candidates to create and manage their personal profile.
FR-05	The system shall allow candidates to update their educational qualifications, skills, projects, certifications, and work experience.
FR-06	The system shall allow candidates to upload resumes in PDF or DOCX format.
FR-07	The system shall store uploaded resumes securely in cloud storage.
FR-08	The system shall allow candidates to replace or delete previously uploaded resumes.

3.1.2 Recruiter Registration and Profile Management

ID	Requirement
FR-09	The system shall allow recruiters to register using company details and official email addresses.
FR-10	The system shall authenticate recruiter accounts before granting access.

FR-11	The system shall allow recruiters to update company profile information.
FR-12	The system shall provide recruiters with a personalized dashboard for managing recruitment activities.

3.1.3 Administrator Management

ID	Requirement
FR-13	The system shall allow administrators to manage candidate accounts.
FR-14	The system shall allow administrators to manage recruiter accounts.
FR-15	The system shall allow administrators to monitor all job postings
FR-16	The system shall generate reports on placement activities and user statistics.

3.2 Resume Management

3.2.1 Resume Upload and Parsing

ID	Requirement
FR-17	The system shall accept resumes in PDF and DOCX formats.
FR-18	The system shall extract candidate information automatically from uploaded resumes.
FR-19	The system shall identify education, skills, experience, projects, and certifications from resumes.
FR-20	The system shall store parsed resume information in the database.

3.2.2 ATS Resume Evaluation

ID	Requirement
FR-21	The system shall generate an ATS score for each uploaded resume.
FR-22	The system shall evaluate resumes based on formatting, skills, education, experience, and keywords.
FR-23	The system shall provide suggestions to improve ATS compatibility.

FR-24	The system shall highlight missing keywords required for a selected job description.
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3.2.3 Resume and Job Description Matching

ID	Requirement
FR-25	The system shall compare uploaded resumes with selected job descriptions.
FR-26	The system shall calculate a Resume Match Score using AI.
FR-27	The system shall identify missing skills required for the job.
FR-28	The system shall recommend improvements to increase the match score.

3.3 Job Management

3.3.1 Job Posting

ID	Requirement
FR-29	The system shall allow recruiters to create and publish job postings.
FR-30	The system shall allow recruiters to edit or remove job postings.
FR-31	The system shall store job descriptions in the database.
FR-32	The system shall categorize jobs based on skills, location, and experience level.

3.3.2 Job Search and Application

ID	Requirement
FR-33	The system shall allow candidates to browse available job opportunities.
FR-34	The system shall provide search and filter options based on skills, location, and company.
FR-35	The system shall allow candidates to apply for jobs using uploaded resumes.
FR-36	The system shall maintain the application history for each candidate.

3.4 AI – Based Interview Preparation

3.4.1 Interview Question Generation

ID	Requirement
FR-37	The system shall generate interview questions based on the selected job description.
FR-38	The system shall generate technical, HR, and behavioral interview questions.
FR-39	The system shall generate personalized questions based on candidate skills and experience.

3.4.2 Mock Interview

ID	Requirement
FR-40	The system shall provide an AI-powered mock interview interface.
FR-41	The system shall record candidate responses during the interview session.
FR-42	The system shall provide AI-generated feedback after the interview.
FR-43	The system shall display an overall interview performance score.

3.5 Candidate Screening

3.5.1 AI Candidate Ranking

ID	Requirement
FR-44	The system shall rank applicants according to Resume Match Score.
FR-45	The system shall display ATS scores along with candidate ranking
FR-46	The system shall provide recruiters with AI-generated candidate summaries.
FR-47	The system shall allow recruiters to shortlist or reject candidates.

3.6 Dashboard and Reports

ID	Requirement
FR-48	The system shall provide a dashboard for candidates showing ATS scores, applied jobs, and interview history.
FR-49	The system shall provide a recruiter dashboard displaying job postings, applications, and shortlisted candidates.

FR-50	The system shall provide an administrator dashboard for monitoring users, jobs, and reports.
FR-51	The system shall generate placement and recruitment analytics reports.

4.0. External Interface Requirements

This chapter describes the external interfaces of the **AI-Powered Resume Screening and Personalized Preparation Platform**. It specifies how users interact with the system and how the system communicates with external software services and hardware components.

4.1 User Interfaces

The platform provides a responsive, user-friendly, and role-based web interface for Candidates, Recruiters, and Administrators. The interface is designed to ensure easy navigation, accessibility, and efficient interaction across desktops, laptops, tablets, and mobile devices.

The system includes the following user interfaces:

Candidate Interface

- User Registration and Login
- Candidate Dashboard
- Resume Upload and Management
- AI Resume Analysis
- ATS Score Dashboard

- Resume–Job Match Results
- Job Search and Application
- AI Interview Preparation
- Mock Interview
- Profile Management

Recruiter Interface

- Recruiter Registration and Login
- Company Profile Management
- Recruiter Dashboard
- Job Posting and Management
- Candidate Screening
- Candidate Ranking
- Shortlisting and Rejection

Administrator Interface

- Administrator Login
- User Management
- Recruiter Management
- Job Monitoring
- Reports and Analytics
- System Configuration

4.2 Hardware Interfaces

The proposed system does not require any specialized hardware. It operates using standard computing devices with internet connectivity. Minimum hardware requirements include:

- Desktop Computer or Laptop
- Smartphone or Tablet
- Keyboard and Mouse
- Internet Connection

Recommended system configuration:

Component	Specification
Processor	Intel Core i3 / AMD Ryzen 3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	2 GB available space
Internet	Broadband Connection

4.3 Software Interfaces

The system communicates with several software components and third-party services to provide AI-powered functionalities.

Software	Purpose
React.js	Frontend development
FastAPI	Backend REST APIs
PostgreSQL	Database Management

SQLAlchemy	Database ORM
Google Gemini API	Resume analysis, ATS scoring, interview question generation
Clouinary	Resume file storage
JWT	User Authentication
Tailwind CSS	User Interface Styling

4.4 Communication Interfaces

The system communicates over the Internet using secure communication protocols.

Communication specifications include:

- HTTP/HTTPS protocol for client-server communication.
- RESTful APIs for frontend-backend communication.
- JSON as the standard data exchange format.
- Secure authentication using JWT tokens.
- Cloud-based communication with Google Gemini API for AI functionalities.
- Secure communication with Clouinary for resume storage.

5.0. Non-Functional Requirements

This chapter describes the quality attributes and operational characteristics of the AI-Powered Resume Screening and Personalized Preparation Platform. These requirements define how the system should perform and ensure reliability, security, usability, maintainability, and scalability.

5.1 Performance Requirements

The system shall provide satisfactory performance under normal operating conditions.

5.1.1 Response Time

ID	REQUIREMENT
NFR – 01.	The system shall respond to user requests within 5 seconds under normal operating conditions.
NFR – 02.	Resume upload shall be completed within 15 seconds, depending on file size and network speed.
NFR – 03.	AI-generated resume analysis and ATS score shall be displayed within 15–30 seconds.
NFR – 04.	Job search results shall be displayed within 7 seconds.

5.1.2 Scalability

ID	REQUIREMENT
NFR – 05.	The system shall support multiple users accessing the application simultaneously.
NFR – 06.	The architecture shall support future expansion by adding new AI modules and additional user roles without significant modifications.
NFR – 07.	The database shall support increasing volumes of resumes, job postings, and application records.

5.1.3 Availability

ID	REQUIREMENT
NFR – 08.	The system shall be available 24 hours a day, 7 days a week, except during scheduled maintenance.
NFR – 09.	Scheduled maintenance activities shall be communicated to users in advance.

5.2 Security Requirements

The platform shall ensure the confidentiality, integrity, and availability of user data.

5.2.1 Authentication and Authorization

ID	REQUIREMENT
NFR – 10.	The system shall implement secure user authentication using JWT-based authentication.
NFR – 11.	Role-Based Access Control (RBAC) shall restrict access based on Candidate, Recruiter, and Administrator roles.
NFR – 12.	Passwords shall be securely stored using secure one-way hashing techniques.

5.2.2 Data Protection

ID	REQUIREMENT
NFR – 13.	All communication between the client and server shall use HTTPS.

NFR – 14.	Uploaded resumes and user information shall be securely stored.
NFR – 15.	Only authorized users shall access personal and recruitment data.

5.3 Reliability

ID	REQUIREMENT
NFR – 16.	The system shall provide consistent and accurate results during resume analysis and job matching.
NFR – 17.	The application shall recover gracefully from unexpected failures without data loss.
NFR – 18.	Error messages shall be displayed clearly without exposing sensitive system information.

5.4 Usability

ID	REQUIREMENT
NFR – 19.	The system shall provide a simple and intuitive user interface.
NFR – 20.	The application shall support responsive design for desktops, tablets, and mobile devices.
NFR – 21.	Navigation shall be consistent across all modules.
NFR – 22.	Users shall be able to complete major tasks with minimal training.

5.5 Maintainability

ID	REQUIREMENT
NFR – 23.	The application shall follow a modular software architecture.
NFR – 24.	Source code shall be properly documented and organized.
NFR – 25.	New features shall be added without affecting existing functionalities.
NFR – 26.	APIs shall be designed following RESTful standards.

5.6 Compatibility

ID	REQUIREMENT
NFR – 27.	The system shall support modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.
NFR – 28.	The application shall be compatible with Windows, Linux, Android, and macOS through a web browser.

5.7 Backup and Recovery

ID	REQUIREMENT
NFR – 29.	The database shall be backed up periodically.
NFR – 30.	The system shall restore application data in case of unexpected failures.
NFR – 31.	Cloud-based storage services shall ensure secure storage of uploaded resumes.

5.8 Operational Requirements

ID	REQUIREMENT
NFR – 32.	The application shall be deployed on cloud platforms.
NFR – 33.	The system shall support continuous monitoring and logging.
NFR – 34.	The application shall be accessible through a stable internet connection.
NFR – 35.	Future updates shall be deployed with minimal service interruption.

5.9 Legal and Privacy Requirements

ID	REQUIREMENT
NFR – 36.	User information shall be collected only for recruitment purposes.
NFR – 37.	Personal information shall not be shared with unauthorized users.

5.10 AI Model Requirements

ID	REQUIREMENT
NFR – 38	The AI module shall generate ATS scores based on resume content and job requirements.

NFR – 39	The system shall generate personalized interview questions using the uploaded job description.
NFR – 40	AI-generated recommendations shall be presented within 20 seconds under normal conditions.
NFR – 41	The AI module shall produce consistent outputs for similar inputs and provide meaningful suggestions to improve resumes and interview preparation.

6.0. Other Requirements

6.1 Deployment Requirements

The AI-Powered Resume Screening and Personalized Preparation Platform shall be deployed as a cloud-based web application to ensure accessibility, scalability, and reliability. The frontend application will be deployed using Vercel, while the backend services will be hosted on Render. The application database will be managed using Neon PostgreSQL, and uploaded resumes will be securely stored using Cloudinary. The deployment architecture shall support secure communication between all components using HTTPS.

6.2 Data Management Requirements

The system shall securely manage all user and application data throughout its lifecycle. Candidate profiles, recruiter information, resumes, job postings,

applications, ATS scores, and interview records shall be stored in a relational database. Uploaded resumes shall be validated before storage, and duplicate or invalid records shall be handled appropriately. Regular database backups shall be maintained to ensure data availability and recovery in case of failures.

6.3 AI Model Requirements

The Artificial Intelligence module shall support resume parsing, ATS score generation, resume and job description matching, skill gap analysis, and personalized interview question generation. The AI module shall provide consistent and relevant outputs based on the candidate's profile and the selected job description. AI-generated recommendations shall assist both candidates and recruiters in making informed decisions while maintaining acceptable response times.

6.4 Training and User Support

The system shall provide sufficient documentation and guidance for all users. Students, recruiters, and administrators shall have access to user guides, FAQs, and help pages to understand system functionalities. The application shall include informative error messages and user-friendly interfaces to minimize the need for external technical support.

6.5 Future Enhancements

The proposed system is designed with a modular architecture to support future enhancements without affecting the existing functionalities. Future improvements may include:

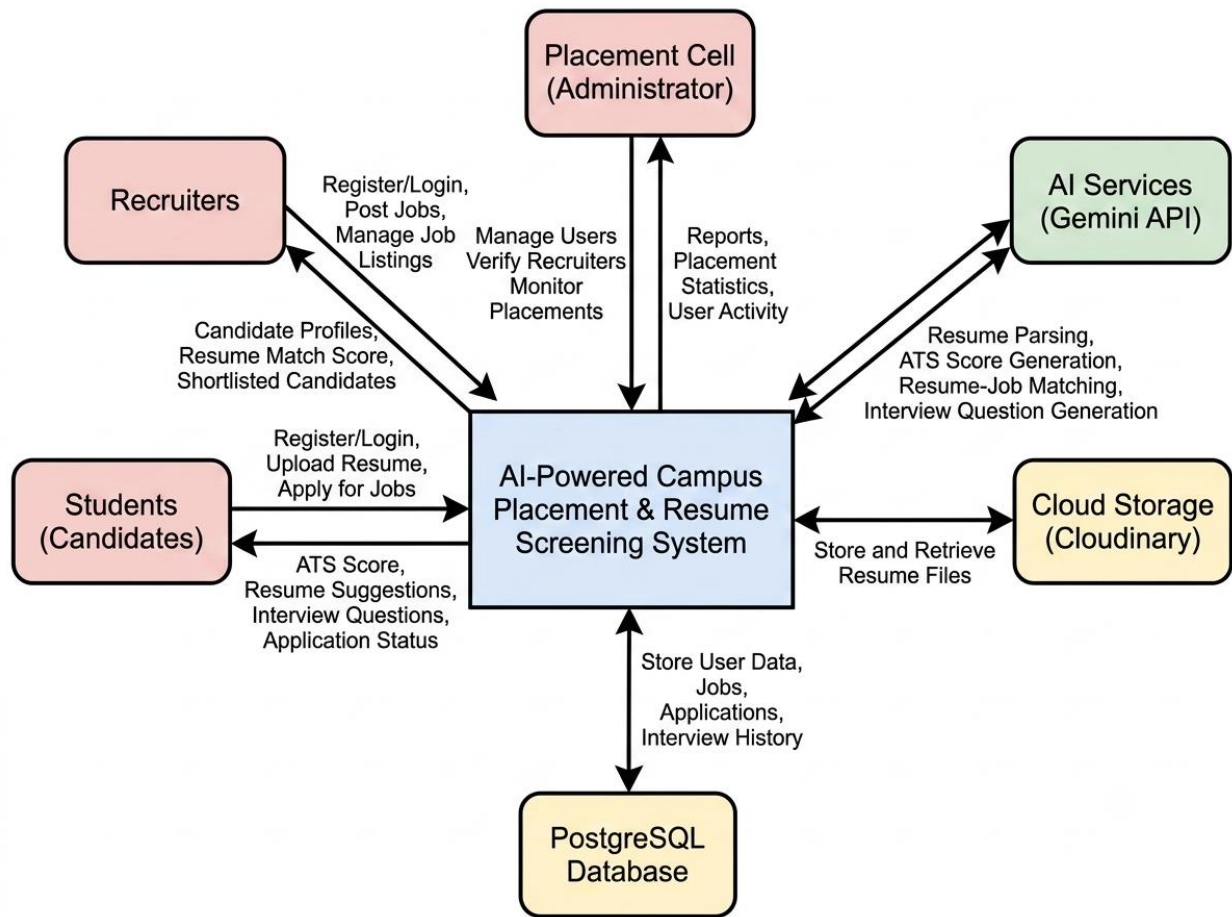
- Integration with LinkedIn and GitHub profiles.
- AI-powered video mock interviews.
- Coding assessment integration.
- Real-time interview scheduling.
- Email and SMS notifications.
- Mobile application support.
- Integration with external job portals.
- Advanced analytics and placement reports.

6.6 Appendix A: Analysis Models

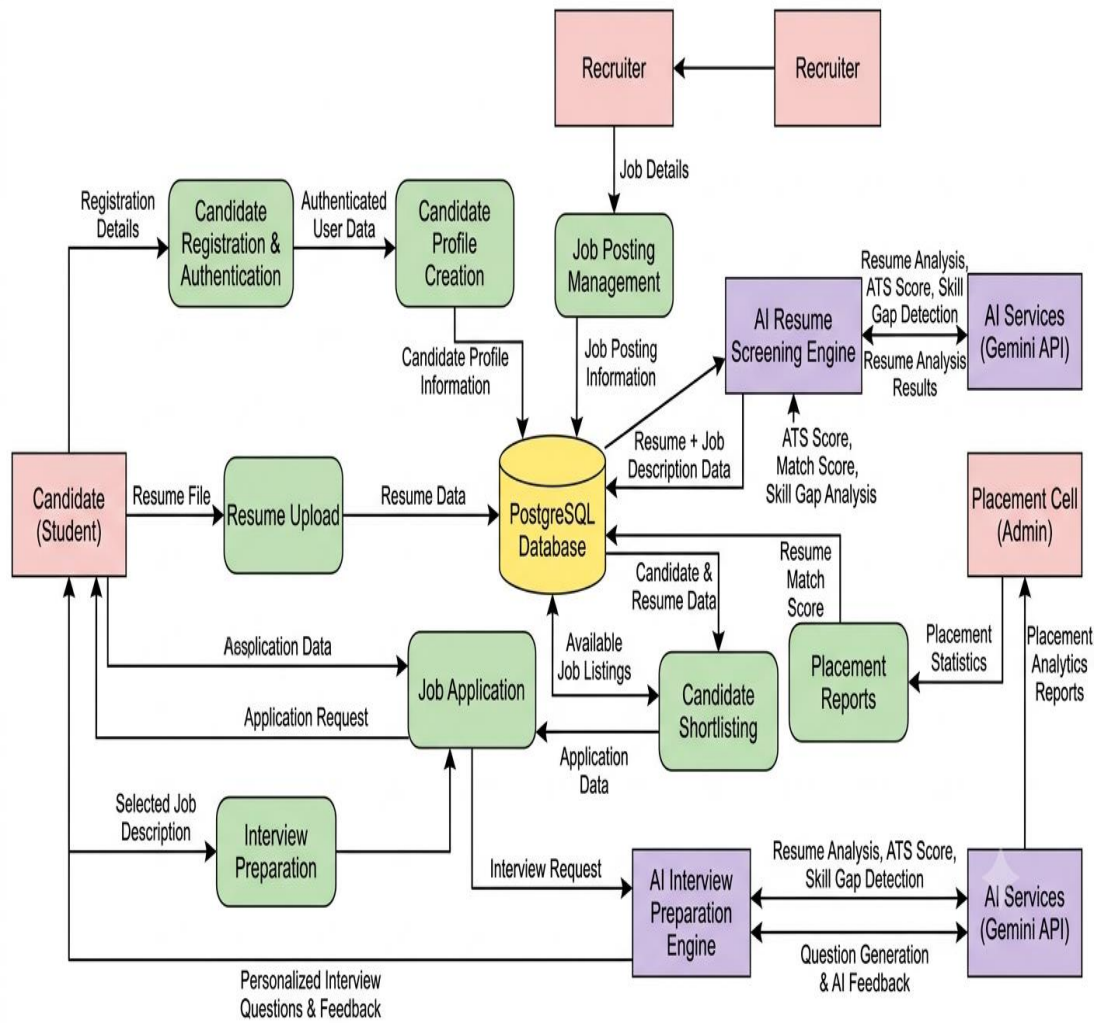
This appendix will include the following analysis models:

1) Context Diagram

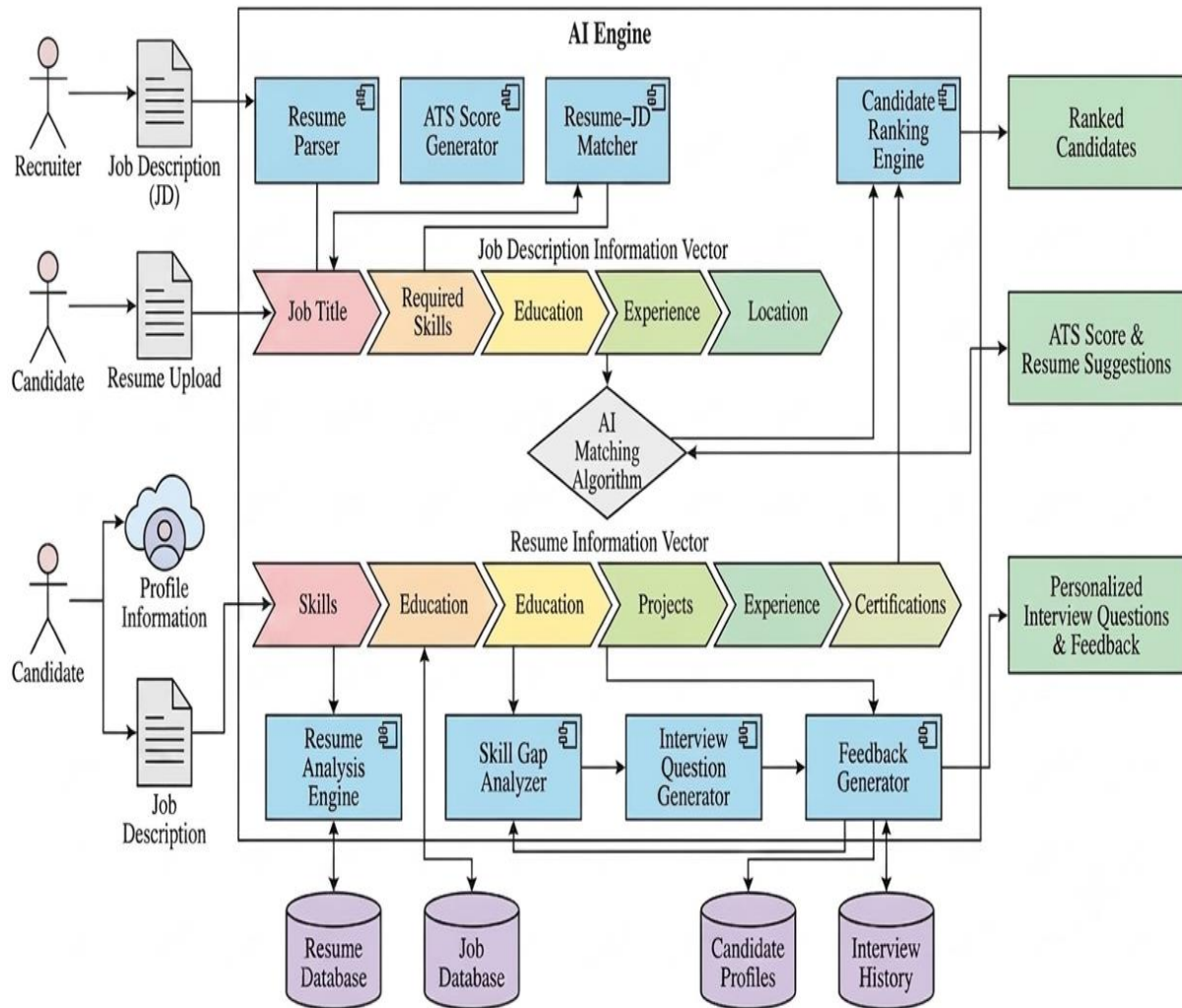
- System boundaries and external entities (Level 0)



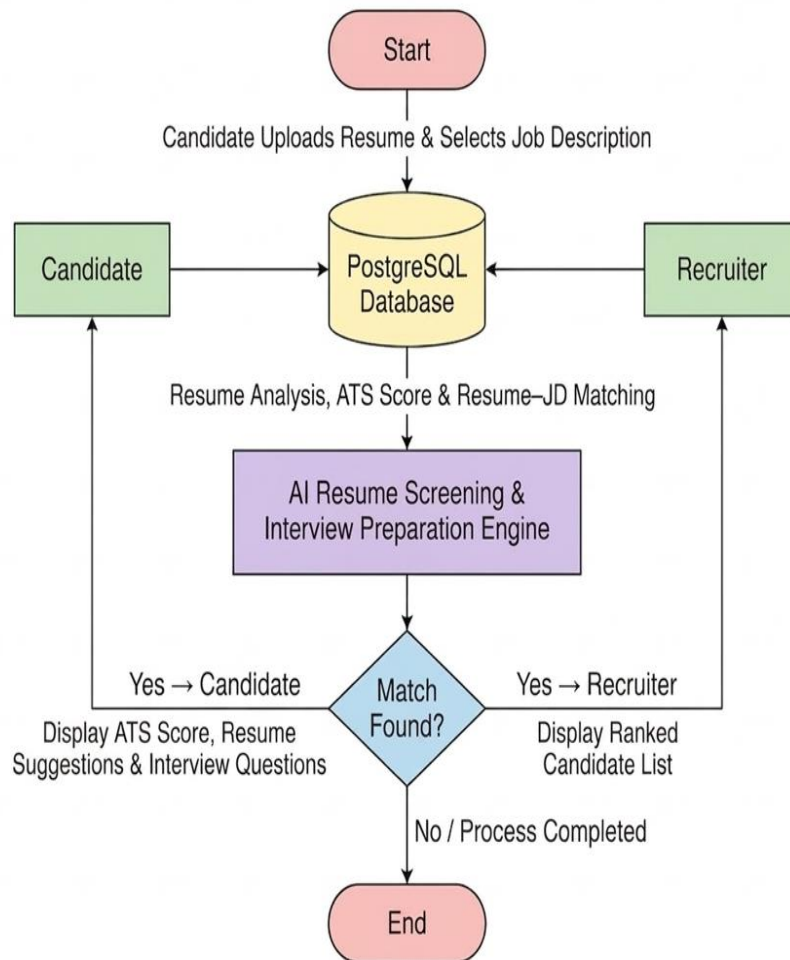
- Data flows between the system and external entities (Level 1)



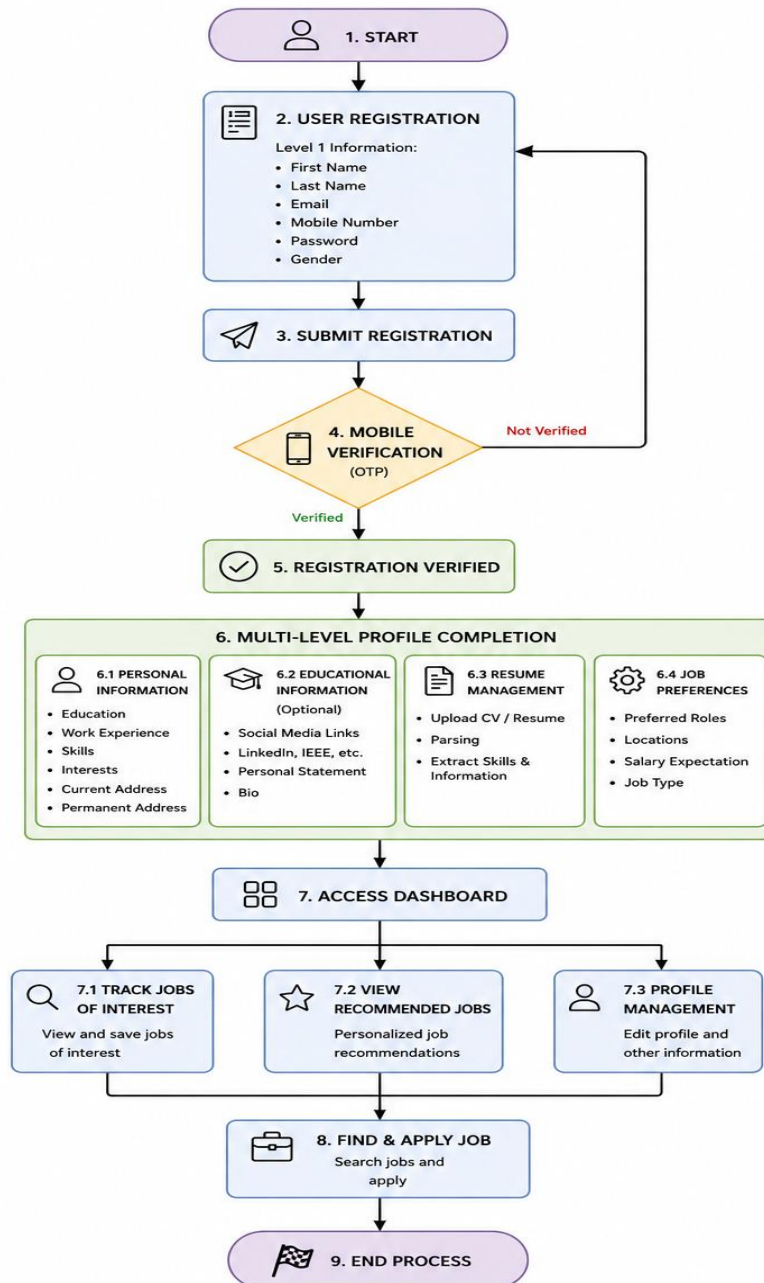
2) AI Engine Component Diagram



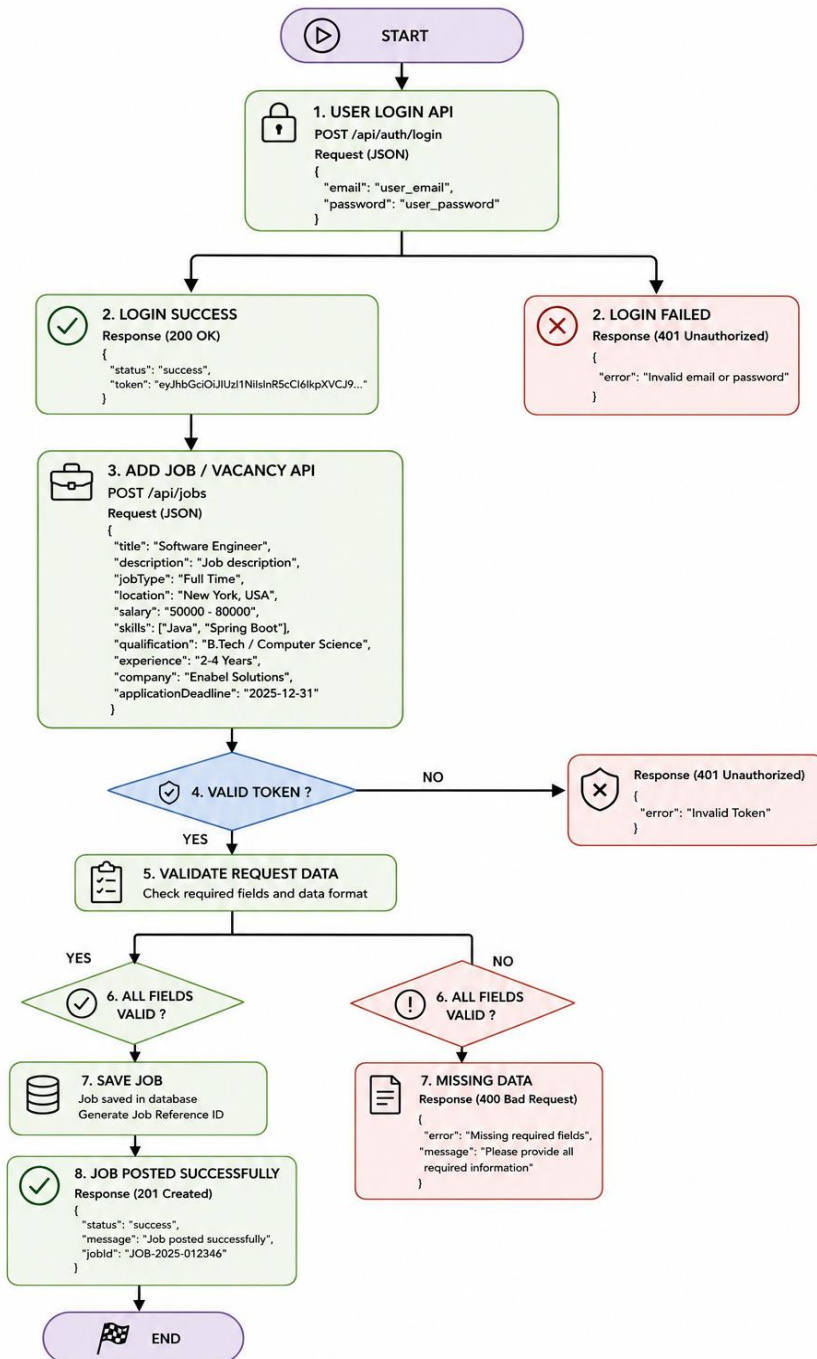
3) Process Flow Diagrams



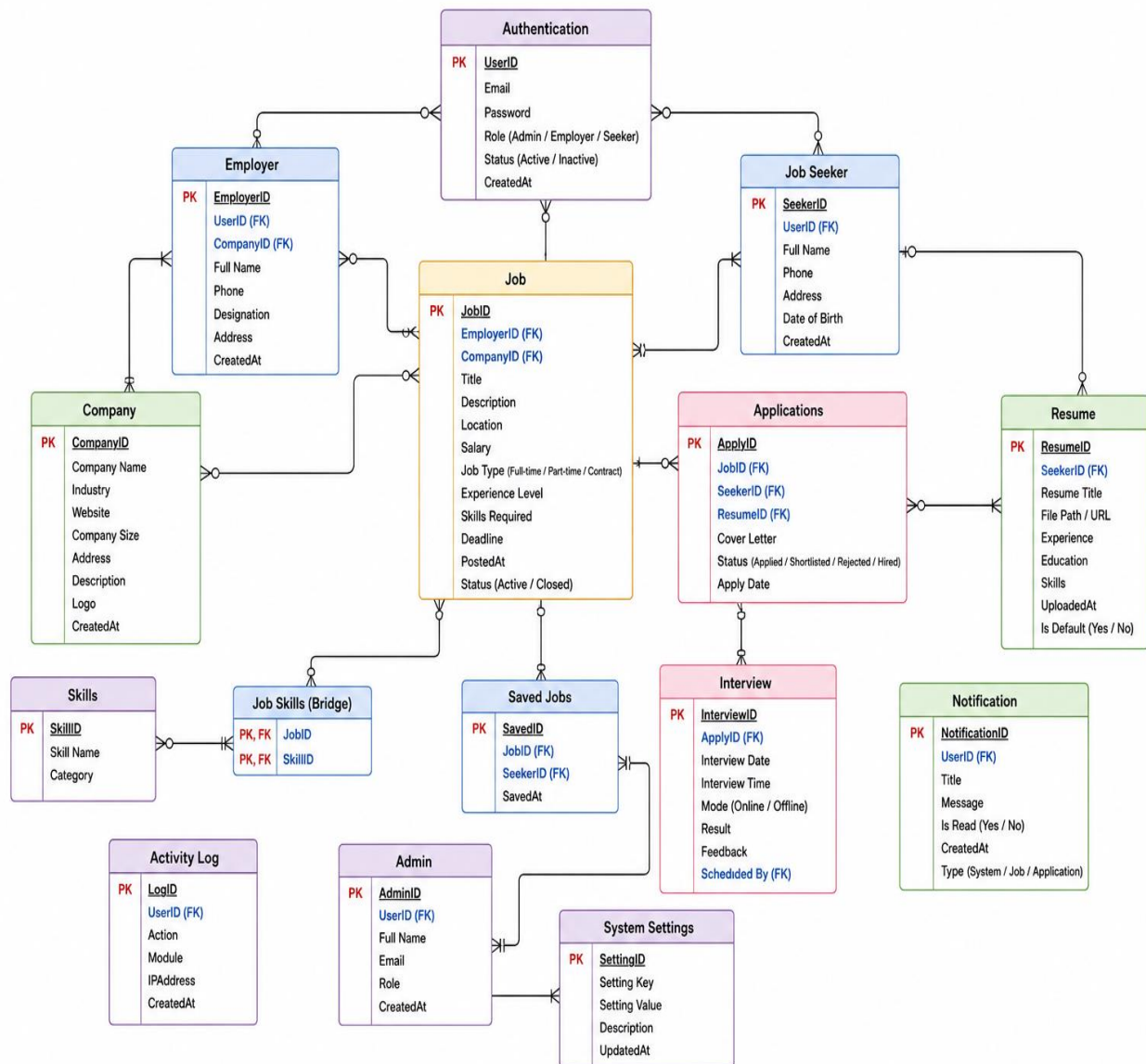
- Job seeker registration and profile management



- Employee Registration and Profile management



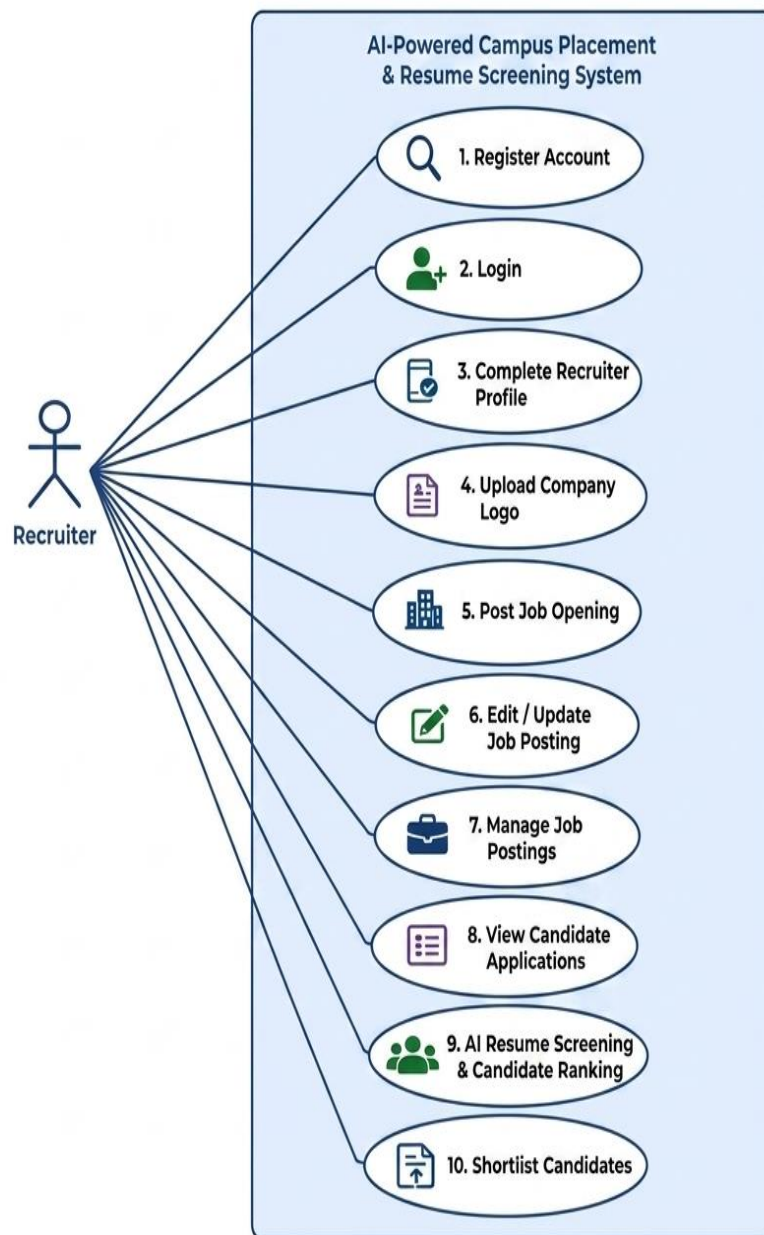
4) Entity Relationship Diagram



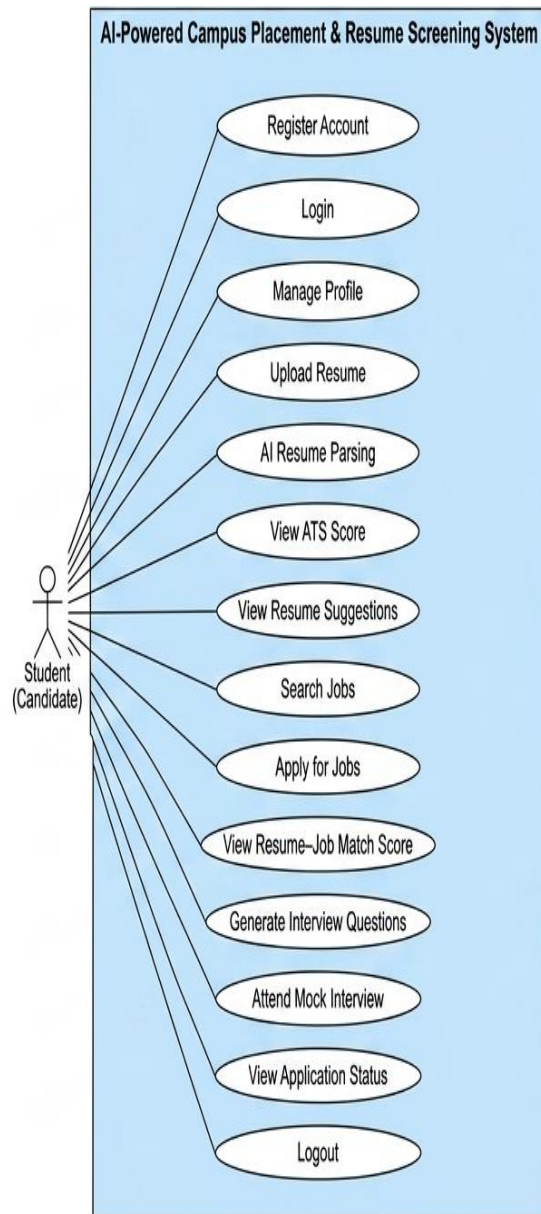
5) Use Case Diagrams

Key system use-cases:

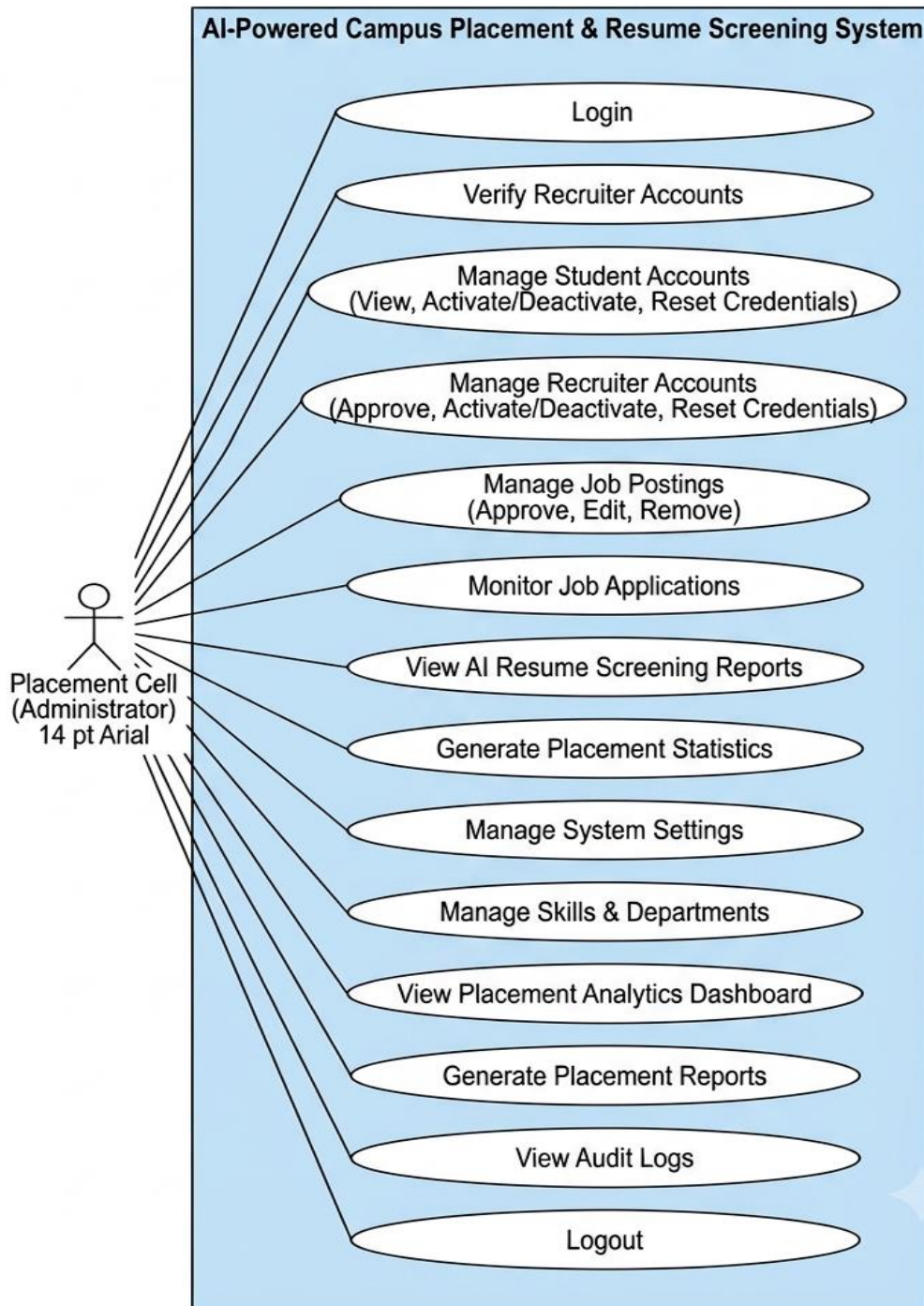
- Recruiter use case diagram



- Student use case diagram



- Placement cell administrator use case diagram:



6.6 Appendix B: Issues List

Appendix B lists the current limitations, pending features, and future improvements identified during the development of the AI-Powered Resume Screening and Personalized Preparation Platform. These items are planned for future iterations and do not affect the core functionality of the current release.

ISSUE ID	DESCRIPTION	STATUS
IS – 01.	Integration with LinkedIn profile import	Future Work
IS – 02.	AI – powered video mock interview	Future Work
IS – 03.	Email and SMS notification service	Planned
IS – 04.	Coding assessment integration	Planned
IS – 05.	Multi-language support	Future Work
IS – 06.	Mobile application development	Future Work